

James Golden

AI Engineer
Software Engineer
Full Stack Developer

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EDUCATION

Apr 2014 ~
May 2017

EDUCATION

Kwantlen Polytechnic University (KPU), Richmond, BC

Graduated: May 2017

Master's Degree in Computer Science

- Relevant Coursework: Custom CRM Platforms, AI Platforms, Social Applications for Cross-Platforms
- Achievements: Developed an AI-driven chatbot project focused on enhancing customer service experiences, inspired by OpenAI technologies; Published a paper on AI ethics and applications at the annual AI and Machine Learning Conference.

SUMMARY

I'm a Full Stack Developer and AI Engineer with extensive experience in creating innovative AI solutions and applications. My work primarily focuses on developing AI-driven products that are both cutting-edge and accessible, with a particular emphasis on AI agent development. I've led and contributed to projects across various industries, such as HR services, recruitment, and content generation, using my expertise to boost operational efficiency and user engagement.

Throughout my career, I've worked with a wide range of programming languages and frameworks, including JavaScript, Python, Java, React, Node.js, and Spring Boot, to build robust and scalable applications. I have a proven track record of integrating advanced AI technologies like machine learning, natural language processing, and computer vision, developing sophisticated AI agents that handle complex tasks and interactions. My experience also extends to developing AI-driven voice agents for VoIP systems to streamline communication processes and enhance client interactions.

I'm skilled in deploying and managing applications on cloud platforms like AWS, GCP, and Azure AI, and I use DevOps tools such as Docker, Kubernetes, and GitHub Actions to ensure seamless integration and efficient operation. My knowledge of databases, including PostgreSQL, MongoDB, and DynamoDB, ensures strong data management and retrieval capabilities.

Beyond my technical skills, I excel at leading cross-functional teams and managing projects with tools like Jira, Trello, and Asana, ensuring timely delivery and high-quality outcomes. I have a solid background in developing RESTful APIs and employing WebSockets for real-time communication, enhancing the interactivity and responsiveness of applications.

I'm committed to continuous learning and staying updated with the latest advancements in AI and software development, allowing me to bring innovative ideas and solutions to the projects I work on. My goal is to create AI

products that are not only technically advanced but also user-friendly and globally accessible, making a significant impact in the AI technology field. As an individual developer, I'm dedicated to delivering exceptional results and driving innovation in AI agent development.

**PROFESSIONAL
EXPERIENCE**

Aug 2022 ~
Currently

PM & Full Stack Developer & AI Engineer

- March 2024 ~ November 2024 Epic Caller. Panama, US

As a VoIP Engineer and Backend Engineer, I led the development of an AI-driven voice agent system for HR services on a SaaS platform.

We initially used Twilio but switched to FreeSWITCH due to challenges with handling DTMF signals, allowing us to manage over 500 concurrent calls efficiently.

For speech-to-text capabilities, I integrated Deepgram, and later transitioned our text-to-speech services to ElevenLabs to support 28 languages and reduce latency.

To enhance the AI agent, I implemented a Retrieval-Augmented Generation (RAG) system using LangChain and Pinecone, enabling it to process various document types and provide contextually relevant responses.

The AI voice agent achieved a latency of 600ms to 1000ms while handling inputs seamlessly.

Additionally, I developed an advanced order assistant chatbot that could understand customer requests and respond with images and pricing.

Through effective project management and teamwork, I successfully deployed the AI voice agent system, demonstrating its capabilities through live demo calls.

- October 2023 ~ February 2024 Applicantai. NY, US

At Applicant AI, I led a team in developing an innovative AI recruitment model to optimize candidate screening.

We used Apache MXNet to build a deep learning model that improved candidate matching accuracy.

I implemented FastAPI for a high-performance RESTful API on the backend and used PostgreSQL for robust data management.

On the frontend, I developed a dynamic user interface using Svelte.

Additionally, I set up WebSocket connections for real-time communication and used Redis for caching to enhance performance.

As project manager, I coordinated activities and timelines, leading to a successful platform launch that improved recruitment efficiency for clients.

- April 2023 ~ October 2023 Writastic, US

At Writastic, I served as an AI Engineer and Full Stack Developer, where I led the development of a content rewriting model using Python and TensorFlow, significantly enhancing our content generation capabilities.

I designed and implemented the backend infrastructure with Node.js and Express, ensuring efficient data processing and seamless integration with AI models.

On the frontend, I utilized React to create a dynamic user interface that allows users to rewrite content quickly.

I established RESTful APIs for real-time communication between the frontend and backend, while MongoDB was used for scalable data storage.

To optimize application performance, I implemented caching strategies using Redis, which reduced response times during peak usage.
My collaborative efforts with cross-functional teams focused on delivering high-quality software solutions that improved user engagement and productivity on our platform.

- October 2022 ~ April 2023 **Resolve AI. Calgary, Canada.**

During my time as a Full Stack Developer at ResolveAI, I played a key role in the creation and launch of our AI chatbot platform.

I was responsible for developing the front-end with React and the back-end using Node.js, making sure that clients had a smooth and intuitive experience when they set up their chatbots.

Part of my job was to integrate APIs to add features like multilingual support and real-time data processing.

Working closely with our design team, I concentrated on designing responsive layouts and optimizing performance, which led to a noticeable boost in user engagement and satisfaction.

Dec 2021 ~
Aug 2022 **Full Stack Developer**

- April 2022 ~ August 2022 **TeChoices IT Solutions, US**

Technologies used: JavaScript, React/Redux, Node.js, Rest, ES6, Koa, MongoDB, Styled components, Server-side rendering.

Responsibilities:

- Development according to the business requirements.
- Development backend Rest API
- Development new application from scratch
- Integration with RFID Readers - Social network integration
- Payment system & multi-currency integration
- Developing multi-language functionality
- Team leading & code review

- December 2021 ~ March 2022 **GrowThsi, US**

Built an API-driven publishing service handling 17M pageviews monthly, performing at 94% cache efficiency.

Cleared more than 200 performance bottlenecks in Mercurial software for large-scale deployments.

Built out the source control query service that queries information in a scalable, effective, & efficient manner.

Extended the build system to integrate with the Mercurial sparse feature, cutting 47% performance issues.

Rebuilt the Mercurial sparse subsystem to move it from an implementation detail to an end-user friendly tool.

Mentored 10 Junior Python developers, training them to work independently and acquire all adequate skills.

Oct 2019 ~
Dec 2021 **Back End Developer**

- September 2021 ~ December 2021 **Grinteq. NY, US**

Develop and support Node.js frameworks and shared services, including NPM packages and RESTful APIs, to enhance user experience.
Create user-friendly designs through prototypes and wireframes, and ensure scalability with comprehensive documentation and unit tests.
Integrate web services, automate infrastructure, and collaborate with vendors to implement critical tools.
Contribute to enterprise-wide tooling and evaluate new technologies for adoption.

- December 2020 ~ September 2021 BookBeam. Boston, US

Developed RESTful APIs and SOAP using Java/Kotlin with Spring Boot for a web application serving over 1 million monthly users, and built a React frontend with Redux along with a Chrome extension.
Utilized Kafka for data streaming and developed a Scala engine with Spark to handle big data from AWS, enhancing sales on Amazon.
Implemented a multithreading solution to optimize a web scraping engine using Selenium and Python.
Managed backend services with MySQL, NoSQL (MongoDB), and GraphQL, deploying and managing microservices on AWS, with later migration to GCP.
Employed JIRA for bug tracking and used Mockito for unit testing, while also overseeing offshore work and conducting system and regression testing.

- July 2020 ~ September 2020 TulaCo. Santa Monica, US

Implemented AWS containers, Kubernetes, and Docker to support Go implementations and streamline application development, deployment, and scaling.
Collaborated with technical staff and product teams to ensure smooth DevOps workflows and create scalable, secure software solutions.
Automated infrastructure and monitoring services with custom scripts, and developed dashboards for proactive issue resolution.
Created reusable code elements and documentation to support security reviews and enhance project maintainability.

- February 2020 ~ Jun 2020 L3 Technologies

Developed enterprise grade web applications for various clients using latest technologies like NodeJS and other JavaScript technologies.
Developed the code to send the automated emails from NodeJS when an action is performed on front-end Angular view pages.
Worked on Mocha package and used chai to use expect method in the unit test cases results on NodeJS.
Used node mailer package in Node.js to send email using sendMail method.
Worked on Node.js for developed intensive web applications like video streaming sites and single page applications.

- November 2019 ~ March 2020 Eide Bailly LLP

My role was Backend Developer and I've used Nodejs to run Grunt tasks and build the project properly.
Worked on multiple JavaScript frameworks (Nodejs) to augment browser based applications with MVC capability.
Working towards the development and upgradation of MEAN-stack web applications using NodeJS and Express.

Developed responsive web page layouts NodeJS

- January 2019 ~ December 2019 **GrowThsi, US**

Used Lambda function on text object methods Context object properties using Nodejs.

Worked on Node.js middleware framework like Express and Hapi.js and essentially developed micro services and served them through middleware

Experienced in Working on modules like MongoDB and mongoose for database persistence using Node JS to interact with AWS Dynamodb.

Designed the API structures with Nodejs running on Nginx.

Jun 2017 ~
Sep 2019 **Front End Developer**

- March 2017 - February 2018 **Prime Holding JSC**

Created the admin application of the web store in React. * Replaced Redux with Apollo Client for GraphQL against a Ruby back-end. * Created a HOC architecture to reuse common GraphQL functionality for listing and filtering. * Created our own custom UI library with styled components. * Used HOC for form manipulation.

- May 2017 - February 2018 **Proco**

Created the admin application of the web store in React. * Replaced Redux with Apollo Client for GraphQL against a Ruby back-end. * Created a HOC architecture to reuse common GraphQL functionality for listing and filtering. * Created our own custom UI library with styled components. * Used HOC for form manipulation.

- April 2018 - August 2019 **Light Hours**

Developed a restaurant platform where customers and food places can communicate directly using React/Redux-saga, MUI, Google Map API, S3, Node/Express.js, MongoDB, etc. * Improved SEO, accessibility, best practices, and performance up to 95% based on lighthouse score. * Integrated Mailgun for email delivery, tracking, and marketing functionality. * Set up CI/CD using Github pipeline, Wix, Godaddy, S3, EC2, Cloudfront and Nginx etc.

- May 2017 - February 2018 **Xchange Rate**

React & Laravel, Node, MongoDB project. My role was Front End Developer. Built React pages and reusable components with React Bootstrap and Rest APIs with Node Express framework. * Faced very complex trading strategy and large data processing. Coin trading web service. * React-redux, react-saga, react-hooks, rest api, 3 third party apis, Unit test with Jest, Typescript, GraphQL. * Vue & PHP Laravel Developer for developing real estate website. All pages built by re-usable component, Pixel perfect from XD design. * Built REST apis and cronjob for inputting real estate data from 3 third party

**SKILLS &
ABILITIES****Programming Languages & Frameworks**

- JavaScript, TypeScript, Node.js, Nest.js, Express.js
- Python, Django, Flask, FastAPI
- Java, Kotlin, Spring Boot, Webflux
- Swift, Android, Flutter, React Native
- .NET Framework, ECMAScript, Rust, PHP, HTML, CSS, Sass, LeSS Framework

Front-End Development

- React.js, Next.js, Tailwind, Bootstrap
- Vue.js, Angular

Back-End & Databases

- PostgreSQL, MySQL, MongoDB, DynamoDB, Oracle, SQLite
- SQL, GraphQL, JDBC, NoSQL, Apache Nifi
- Sequelize, TypeORM, Hibernate, Spring JPA

Cloud & DevOps

- AWS (S3, EC2, CodeCommit, API Gateway, Lambda, DynamoDB, IAM, CloudWatch, Serverless Framework), GCP, DigitalOcean, Heroku, Azure AI
- Docker, OpenShift, Kubernetes
- GitHub Actions CI/CD, Terraform, Helm

Communication & APIs

- WebSockets, socket.io
- Twilio API, Stripe API, SendGrid API
- Firebase, JWT, Google Sign-In, Auth0, OAuth

AI & Machine Learning

- OpenAI, ChatGPT, Azure AI, Google AI, IBM Watson
- TensorFlow, PyTorch, Keras, Theano
- Scikit-learn, XGBoost, LightGBM, CatBoost
- Hugging Face Transformers, NLTK, SpaCy, OpenCV, TensorFlow.js, ONNX
- MLflow, TensorBoard, Weights & Biases, Apache MXNet, Caffe, CNTK
- Apache Spark MLlib, Amazon SageMaker, Google Cloud AI Platform, Azure Machine Learning, DataRobot, Dataiku

Testing

- Mocha, Jest, JUnit, Mockito, JMeter

VoIP & Communication

- SIP, H.323, RTP, WebRTC
- FreeSWITCH, Asterisk, Kamailio, OpenSIPS
- Twilio, Cisco Unified Communications Manager, Microsoft Teams, Zoom Phone, Avaya
- Grandstream, Yealink, VoIP Gateways (Cisco, Patton, Grandstream)
- IP Phones (Polycom, Cisco, Yealink), PBX Systems (3CX, Elastix, FreePBX)
- VoIP Softphones (Zoiper, Linphone), Cloud VoIP Providers (RingCentral, Vonage, 8x8)

Software Development & Deployment

- FastAPI, Flask, Django for AI model deployment
- Jupyter Notebook, Google Colab for development and experimentation

Industry Applications

- E-commerce, Fintech, Healthcare, Real Estate, E-sport, Software/AI SaaS, Manufacturing

Development Deliverables

- Web Application, Custom PHP, Desktop Application
- Website Redesign, Responsive Design
- Software Debugging, Web Development, Ecommerce Website

Version Control and Collaboration

- Git, GitHub, GitLab, Bitbucket, Subversion (SVN)

Project Management and Issue Tracking

- Jira, Trello, Asana, Monday.com, ClickUp, Microsoft Project

Continuous Integration/Continuous Deployment (CI/CD)

- Jenkins, Travis CI, CircleCI, GitHub Actions, GitLab CI/CD

Collaboration and Communication

- Slack, Microsoft Teams, Zoom, Discord, Telegram, Whatsapp, Confluence
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